

BIG SABLE POINT, MI, USAWMO: **997802**Lat: **44.0550N**Lon: **86.5140W**Elev: **610**StdP: **14.37**Time Zone: **-5.00 (NAE)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.2	(c) 8.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 38.0	(k) 27.4	(l) 35.7	(m) 21.6	(n) 17.2	(o) 280	(p) 0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.0	(c) 78.7	(d) N/A	(e) 77.1	(f) N/A	(g) 75.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 33.7	(b) 29.4	(c) 26.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	24.2	24.5	32.2	42.3	52.6	61.0	67.8	68.0	62.0	51.1	40.2	30.8		
	DBStd	17.13	8.71	7.87	9.12	6.98	7.70	6.17	5.20	4.57	6.12	6.83	7.87	7.57		
	HDD50	3352	801	713	558	249	57	2	0	0	1	68	307	596		
	HDD65	7066	1266	1133	1016	680	395	150	33	23	129	435	745	1061		
	CDD50	2078	0	0	7	19	136	331	551	559	361	101	13	0		
	CDD65	315	0	0	0	0	9	30	118	117	38	3	0	0		
	CDH74	664	0	0	0	0	30	53	291	233	56	1	0	0		
	CDH80	17	0	0	0	0	0	1	9	6	1	0	0	0		
(14) Wind	WSAvg	12.4	16.1	14.7	12.2	12.4	10.8	9.2	9.2	9.6	11.2	13.8	15.3	14.2		
(15) Precipitation	PrecAvg	34.5	2.2	1.7	2.0	3.0	3.4	3.4	3.3	3.2	3.5	3.5	3.0	2.5		
	PrecMax	45.4	4.4	3.1	4.9	5.4	8.1	8.5	6.8	5.6	6.7	8.9	6.9	7.5		
	PrecMin	26.9	0.6	0.7	0.4	0.8	0.8	1.2	0.7	0.9	0.6	0.7	0.7	0.5		
	PrecStd	4.6	1.0	0.7	1.0	1.2	1.7	2.0	1.6	1.1	1.6	2.0	1.6	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	42.5	67.2	68.1	77.6	78.7	81.2	80.5	78.4	72.5	62.2	55.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	42.0	39.3	57.2	61.5	73.2	75.4	78.9	78.3	75.5	68.4	58.3	46.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	39.3	37.0	49.0	56.4	68.7	72.9	77.2	76.6	73.3	65.0	54.1	43.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	36.6	35.1	43.8	52.0	64.2	70.5	75.5	75.1	71.2	61.7	51.1	41.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		9.1	9.3	11.1	11.3	13.4	12.5	9.5	10.0	11.0	10.0	9.3	8.1		
	5% DB	MCDBR	11.3	10.0	16.1	18.4	18.9	15.2	10.0	10.1	11.1	12.9	12.6	10.9		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.310	0.333	0.366	0.396	0.407	0.409	0.400	0.383	0.348	0.316	0.288		
	taud		2.453	2.392	2.404	2.340	2.326	2.336	2.342	2.363	2.394	2.476	2.501	2.499		
	Ebn at Noon		266	280	287	285	278	274	272	271	266	263	255	258		
	Edh at Noon		23	29	33	38	39	39	39	36	33	27	22	20		
	RadAvg		344	622	1116	1502	1816	1992	2026	1718	1341	808	448	275		
(45) All-Sky Solar Radiation	RadStd		43	71	115	140	156	134	89	108	114	94	66	38		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page